

Predicting Skin Pigmentation Class from AlphaGenome-Derived Regulatory Signal: A Detailed Results Report

Scope of this document: this report assembles, in one place, the methodology, thinking process, and real results produced across `01_overview_and_vep.ipynb` through `07_gwas_baseline.ipynb` in this directory. Every number below was pulled directly from the executed notebooks (not reconstructed from memory) — cell IDs are cited so any figure can be traced back to its source cell. The intent is to give you and your colleagues everything needed to decide (a) whether to continue investing in this line of work, and (b) whether the current results, as they stand, are publishable.

Headline answer, stated up front so it doesn't get lost in the details: the pipeline is methodologically careful (held-out evaluation, FDR correction, a real classical-GWAS baseline with proper QC, an interpretability pass, extensive internal caveats already documented notebook-by-notebook), but the central results are currently confounded by a design choice that cannot be fixed by more modeling: **"pigmentation class" in this dataset is a population-membership label (African-ancestry vs. European-ancestry 1000 Genomes populations), not a measured phenotype.** Every downstream number — 99% classification accuracy, thousands of "significant" genes/variants — is consistent with the models picking up general population-structure signal rather than pigmentation-specific biology, and two independent checks in this report actively point that direction (GTEx concordance at chance level; a plain classical GWAS baseline finding comparably strong "significance" using 60-year-old statistics). Section 14 lays out concretely what would need to change before this is a defensible publication claim.

1. Motivation and research question

The question driving this project: can AlphaGenome — a sequence-to-function model that predicts regulatory tracks (RNA-seq, chromatin accessibility, histone marks, etc.) from DNA sequence alone — be used to (a) distinguish individuals by a pigmentation-associated phenotype from their genotype via predicted gene expression, and (b) score which specific non-coding variants in known pigmentation genes are functionally consequential, in a way that agrees with independent evidence (published literature, GTEx eQTLs) and adds something over classical genetic-association methods?

This was pursued as an exploratory, notebook-driven analysis (not a pre-registered study), evolving through several methodological iterations that are documented as they happened, including at least one approach (automated gene-ranking) that was tried and explicitly rejected once it produced biologically incoherent results (Section 3).

1.1 A framing question worth answering before the results: does predicting expression, rather than working with raw genotype, protect this pipeline from population-stratification confounding?

It's tempting to assume that routing genotype through AlphaGenome into a predicted expression signal — conceptually similar to a TWAS (transcriptome-wide association study), which is often pitched as an improvement over plain GWAS — buys some protection from the ancestry-confound problem described in Section 2. It does not, and it's worth being precise about why, since the same logic determines how much weight the rest of this report's results can bear.

What TWAS actually improves on over GWAS has nothing to do with stratification. Classical TWAS (PrediXcan/FUSION-style) trains a sparse predictor of expression from a small set of validated cis-eQTL SNPs in a reference panel, then tests that predicted expression against the phenotype in a separate cohort. Its benefits are statistical and interpretive: fewer tests (one per gene instead of one per SNP), a specific mechanistic hypothesis instead of an anonymous LD block, and sometimes more power when the causal mechanism really is cis-regulatory. **None of that addresses ancestry**

confounding. Standard TWAS practice still requires ancestry PCs / ancestry-matched reference panels for exactly the same reason GWAS does — the statistical-genetics literature on "TWAS confounding" treats LD/ancestry contamination between eQTL weights and the phenotype as an open problem TWAS inherits, not one it fixes.

Mechanistically, why AlphaGenome doesn't change this: predicted expression, however it's computed, is still a deterministic function of local genotype. If genotype in a window is population-differentiated — which pigmentation-gene windows are, almost by definition, since that differentiation is why they were discoverable as pigmentation genes in the first place — then any function of that genotype (linear TWAS weights, or AlphaGenome's nonlinear sequence-to-function model) inherits the same differentiation, unless the function specifically discards the population-correlated component of the input and keeps only the trait-causal component. AlphaGenome has no mechanism for doing that: it has no notion of "this variant is the real pigmentation-causal one" versus "this variant just happens to differ in frequency between African and European populations because of demographic history" — it returns a predicted track for whatever sequence it's given.

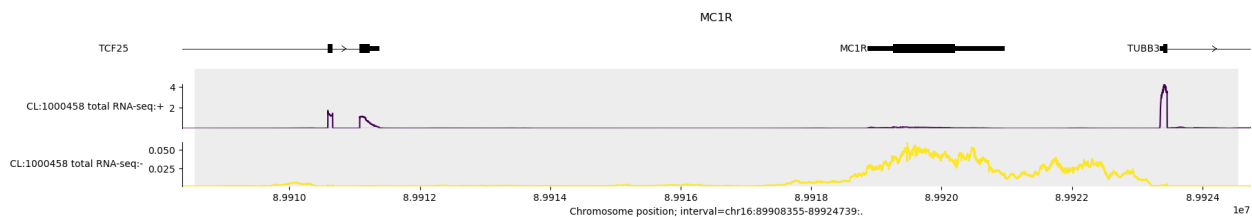
If anything, **this pipeline is more exposed to that leakage than classical TWAS, not less:** real TWAS restricts to a small, cross-validated set of cis-eQTL SNPs, while the two scalar classifiers here (Section 6) run AlphaGenome on an individual's entire real diploid consensus sequence for the window — every variant present, causal or not, shapes the output — and the neural model (also Section 6) goes further still, using the raw per-position signal across ~1M positions rather than a single gene-level expression scalar, an enormous, largely uninterpreted representation with far more surface area for ancestry-correlated (not necessarily biologically causal) structure to leak into than a handful of validated eQTL weights.

Where a sequence-to-function model like AlphaGenome plausibly would offer something GWAS/TWAS structurally cannot — fine-mapping within an LD block, i.e. distinguishing among near-perfectly linked SNPs by recognizing that only one of them sits in a real regulatory motif — is a claim about scoring individual variants, not about classifying individuals, and it is exactly what Section 9's GTEx concordance check tests. That check comes back at chance level (52.4%), so before AlphaGenome can be trusted to adjudicate between candidate causal SNPs within a block, it would first need to reliably get the sign right for eQTLs with independently known measured direction — which it does not yet do reliably on this gene panel. Sections 6–11 should be read with this in mind: nothing about routing genotype through AlphaGenome's expression predictions structurally protects this project's results from the same population-stratification confound that would affect a plain GWAS on the same cohort/phenotype.

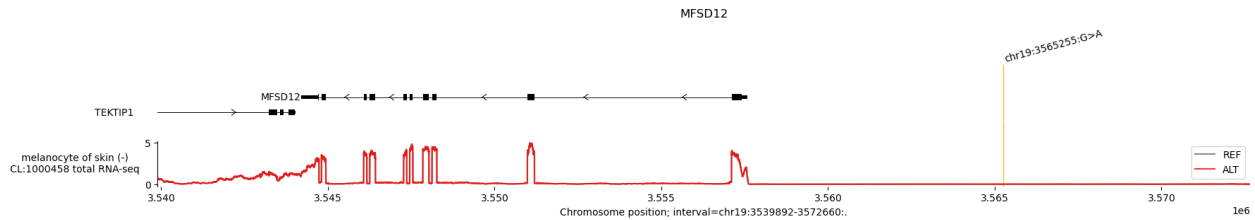
1.2 How AlphaGenome works and how it's used here: a visual walkthrough (notebooks 1–3)

Before the results, it's worth showing concretely what AlphaGenome outputs and the three progressively more involved ways this project uses it — reference-sequence prediction, single-variant effect scoring, and per-individual diploid consensus prediction aggregated across a population. Every image below is a real, unmodified output from the notebooks.

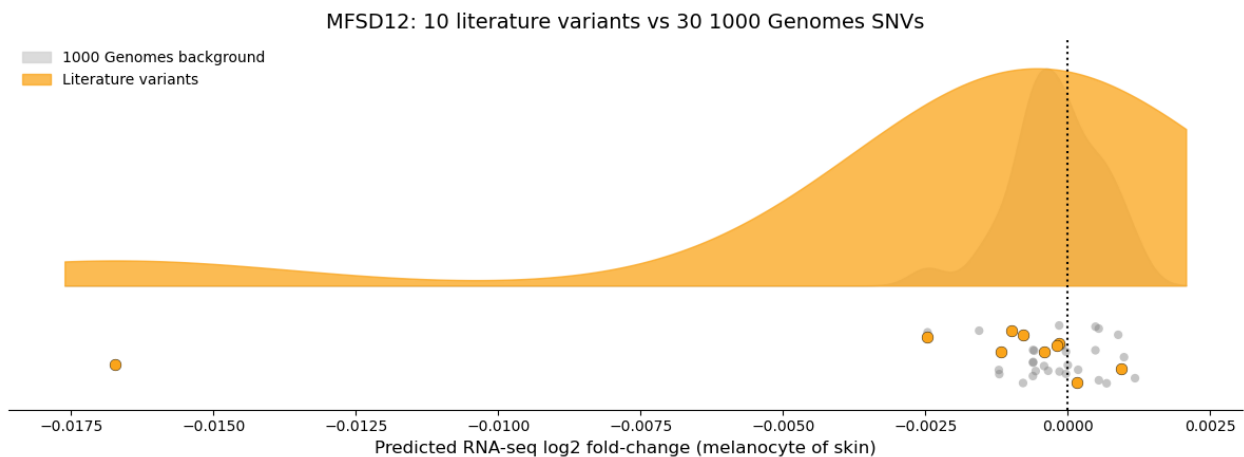
1. What AlphaGenome predicts, given a reference genome window. For any genomic interval, AlphaGenome returns a continuous, per-base-pair predicted signal track for a requested output type (here, RNA-seq) and biosample/ontology term. Below: MC1R's window, unmodified GRCh38 reference, predicted melanocyte-of-skin RNA-seq on both strands, with the gene/transcript structure from GENCODE drawn above the track for reference — this is the basic unit everything else in this project is built from (01_overview_and_vep.ipynb).



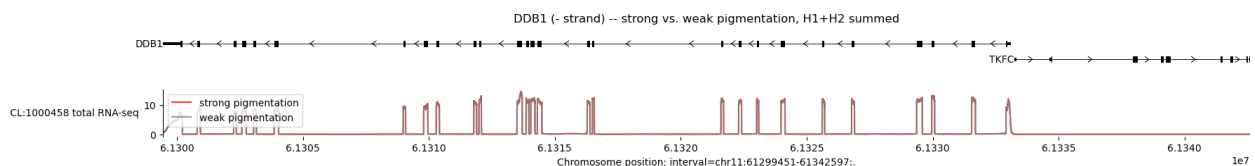
2. Scoring a single variant's predicted effect. AlphaGenome can predict the same track for a mutated sequence (one allele substituted at one position) and compare it against the reference prediction — the difference is the model's predicted regulatory consequence of that specific variant. Below: one of MFSD12's literature (GTEx eQTL) variants, chr19:3565255:G>A, REF vs. ALT predicted tracks overlaid (they are visually close here — consistent with the generally small per-variant effect sizes reported throughout this project, e.g. Section 5's fold-change table).



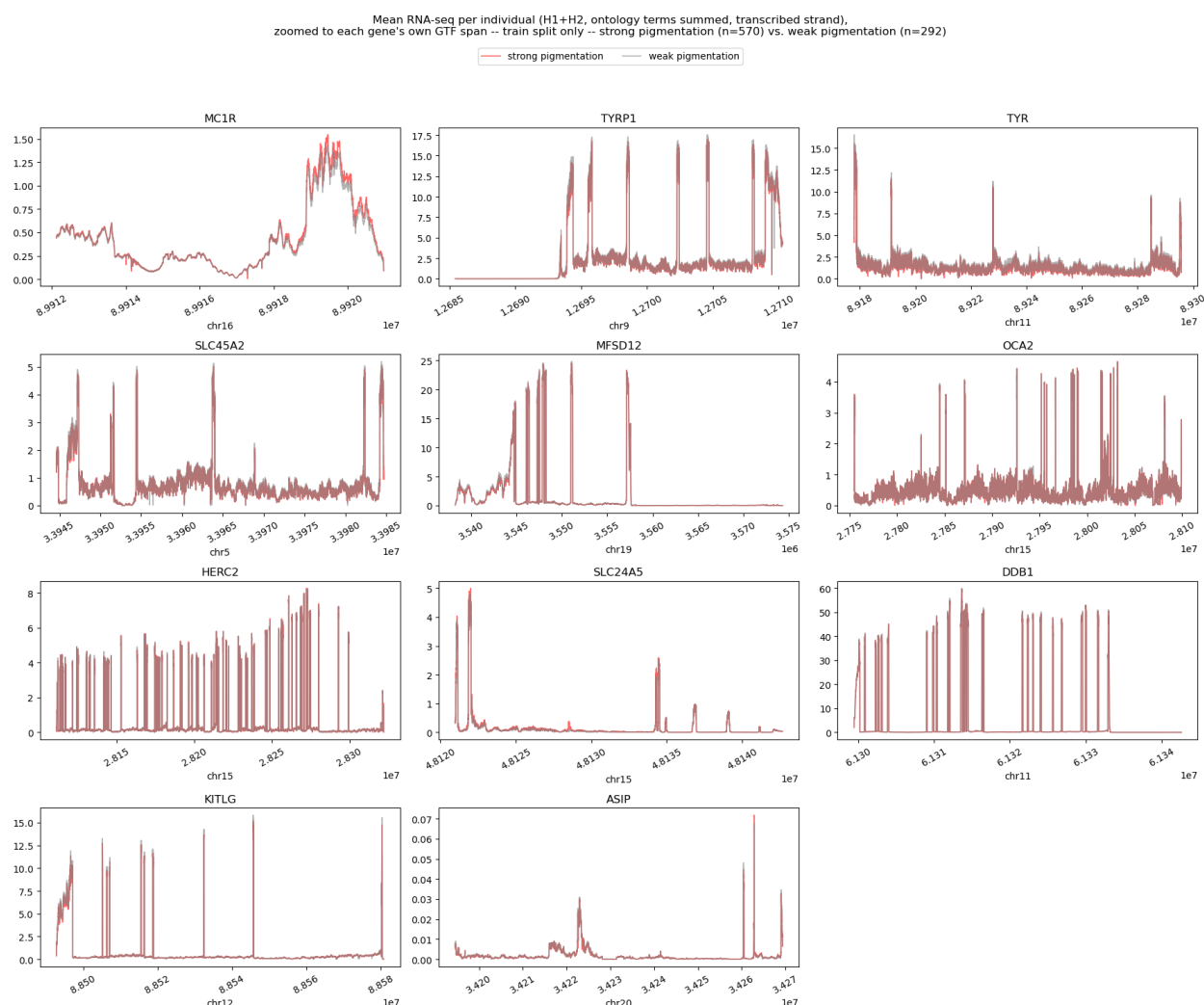
Individual variant scores are aggregated and compared against a random background set of real 1000-Genomes variants from the same window, to check whether curated/literature variants stand out from ordinary genetic variation (this specific comparison method, and its role in Sections 8–9, is described in 01_overview_and_vep.ipynb):



3. Per-individual, diploid, personalized prediction. Real individuals are diploid and carry many variants at once, not one substitution in isolation. For each person, both haplotypes (H1/H2) have their own consensus sequence built (via bcftools consensus against that person's phased 1000 Genomes genotype) and are each run through AlphaGenome separately, then summed. Below: two specific individuals — one from GWD (African, "strong pigmentation" class) and one from GBR (European, "weak pigmentation" class) — compared at DDB1 (02_individual_predictions.ipynb). Note how close the two individuals' predicted tracks are to each other here — a single pairwise comparison like this rarely shows a clean class difference, which is exactly why the analysis needs to move to the population level next.



4. From one pair of individuals to a population-level comparison. 03_population_analysis.ipynb runs this same per-individual diploid prediction for all 862 train-split individuals and averages the result within each class, for every one of the 11 genes at once. This is the figure that directly motivates Section 5's per-gene statistical test below — look closely at how similar the red ("strong pigmentation") and gray ("weak pigmentation") curves are for every gene, despite most of these differences later reaching extremely small p-values purely because of the large sample size (n=862):



2. Data: cohort, phenotype definition, and its central limitation

Source: 1000 Genomes Project, high-coverage (30x) phased SNV/INDEL/SV panel, GRCh38. 1,072 individuals across 8 populations were used, split into two classes by population membership (`notebooks/data/experiment.json` , `class_map`):

Class	Populations	N
"strong pigmentation"	YRI, ESN, LWK, MSL, GWD (African)	703
"weak pigmentation"	FIN, CEU, GBR (European)	369

This is the single most consequential methodological decision in the whole project, and it is worth stating plainly: there is no measured pigmentation phenotype anywhere in this pipeline (no melanin index, no reflectometry, no self-report). "Strong"/"weak pigmentation" is a coarse proxy built entirely from which 1000 Genomes population panel an individual belongs to. This is a reasonable, common simplification in population-genetics teaching examples (these population groups do differ substantially in average skin pigmentation, and the panel of genes chosen — Section 3 — is genuinely pigmentation-associated in the literature), but it means:

- The "label" being predicted is inseparable from all African-vs-European population structure — not just pigmentation-relevant structure. Any regulatory or genotypic difference between these populations (drift, other selected loci, sequencing/reference-bias artifacts) is a candidate confound for every result below.

- A model can achieve very high accuracy at this task by detecting "does this genotype look African or European" without learning anything specific to pigmentation biology, because these specific gene regions are themselves among the most population-differentiated loci in the genome (that is why they were discoverable as pigmentation genes in the first place — but it also means they are unusually easy to classify by ancestry alone, independent of any AlphaGenome insight).

Every subsequent result needs to be read through this lens. Section 14 returns to this as the primary blocker to publication in current form.

Family structure and the train/test split: 1000 Genomes includes trios and multi-member families — 214 of the 588 families in this dataset have more than one member (698 of 1,072 individuals). A naive random split would put relatives on both sides of train/test and leak genotype (and thus predicted expression) across the boundary. `family_aware_train_test_split` (`notebooks/utils/family_split.py`) groups by `family_id` within each population (so the population mix is preserved on both sides, not left to chance) and splits whole family groups, 80/20:

Class	Total	Train	Test
strong pigmentation	703	570	133
weak pigmentation	369	292	77
Total	1,072	862	210

This split is reused identically by every notebook (fixed seed), so "held-out" numbers throughout this report are on the same genuinely-unseen 210 individuals.

3. Gene panel: how the 11 candidate genes were chosen

Two approaches were tried, in this order.

Attempt 1 (rejected): an automated pipeline that (a) queried the Open Targets Platform for any gene with any annotated relationship to "skin pigmentation," then (b) ranked those candidates by raw GTEx skin-eQTL count, taking the top 10. This produced a biologically incoherent list — genes like LRRK2 (Parkinson's disease) and EPM2A (Lafora disease) ranked above OCA2, TYR, and SLC45A2. The root cause: GTEx's per-gene "significant eQTL" count is not LD-pruned to independent causal signals, so genes sitting in large linkage-disequilibrium blocks accumulate large eQTL counts purely from correlated neighboring SNPs, unrelated to true biological relevance. This was diagnosed directly (not assumed) by inspecting the actual candidate ranking, and the whole approach — including its dedicated exploratory notebook and helper module — was deleted once the result was recognized as unsound.

Attempt 2 (used): manual curation from the published skin-pigmentation genetics literature, preferring genome-wide-significant, replicated loci with independent functional support:

Gene	Evidence
MC1R	Valverde et al. 1995 — loss-of-function → red hair/fair skin/UV sensitivity; the most functionally characterized pigmentation gene
TYR	Rate-limiting melanogenesis enzyme; repeatedly replicated GWAS hits (Sturm 2009 review)
TYRP1	Melanogenesis enzyme; Sulem et al. 2008 (Icelandic GWAS), Kenny et al. 2012 (Solomon Islands blond hair)
OCA2	Duffy et al. 2007 — oculocutaneous albinism II gene; one of the two largest-effect eye/skin loci
HERC2	Sturm et al. 2008, Eiberg et al. 2008 — intronic regulatory element controlling OCA2 expression
SLC45A2	Graf et al. 2005 — major light-skin allele in Europeans
SLC24A5	Lamason et al. 2005 — single largest-effect locus for light skin pigmentation, near-fixed in Europeans
MFSD12	Adhikari et al. 2019 (Nat Commun); Crawford et al. 2017 (Science) — melanosome pH regulator, major finding from African/Latin-American pigmentation GWAS
DDB1	Crawford et al. 2017 (Science) — major African-ancestry skin-pigmentation locus
KITLG	Miller et al. 2007 (Cell) — melanocyte development/migration, replicated in European and East Asian GWAS
ASIP	Kanetsky et al. 2002 — MC1R antagonist, one of the most consistently replicated pigmentation GWAS hits

Explicitly excluded despite pigmentation-adjacent literature: EDAR and TCHH (primarily hair/tooth morphology, not skin), MITF/IRF4/BNC2/TPCN2 (more hair/eye-specific than skin-specific; AlphaGenome predictions were generated for these during earlier exploration but they were not carried into the final panel).

Known correlation within the panel: OCA2 and HERC2's 524,288bp AlphaGenome prediction windows overlap by ~235kb (45% of each window), since the genes are physically adjacent on chr15 and the well-established HERC2-intron regulatory element controlling OCA2 sits inside that overlap. Both are retained as separate model features, but their expression signals are expected to be correlated, not independent — relevant whenever comparing their individual coefficients (Section 7) or GWAS hits (Section 11).

4. AlphaGenome prediction pipeline

For each of the 11 genes and each of the 1,072 individuals, a **diploid consensus sequence** was built for both haplotypes (H1/H2) via `bcftools consensus` against the individual's phased 1000 Genomes genotype and the gene's 524,288bp reference window, then run through AlphaGenome (`predict_sequence`) requesting the RNA-seq output. This is $\sim 1,072 \times 11 \times 2 \approx 23,600$ AlphaGenome API calls in total; the full set was generated and cached (`notebooks/.cache/predictions/`) before population-level analysis began — confirmed by notebook 3 reporting complete coverage (n=570/292) for every one of the 11 genes on the train split.

RNA-seq predictions were requested for 3 melanocyte/hair-follicle-adjacent cell ontology terms (`CL:1000458` melanocyte of skin, `CL:0000346` hair follicle dermal papilla cell, `CL:2000092` hair follicular keratinocyte); from notebook 3 onward, modeling was restricted to **melanocyte of skin only** (`CL:1000458`) as the single most directly relevant biosample, rather than summing all three.

5. Population-level differential expression (`03_population_analysis.ipynb`)

For each gene, the train-split individuals' H1+H2 predicted RNA-seq signal (melanocyte of skin, strand-matched to the gene) was averaged within each class, first over the gene's full GTF span, then restricted to MANE Select transcript exons only (introns/flanking sequence excluded — a continuous per-base RNA-seq track summed over a full gene span otherwise mixes in a lot of signal that isn't "gene expression" in the spliced-mRNA sense).

Fold changes are small. Whole-gene-span log2 fold-change (strong/weak) ranges from -0.16 (TYR) to +0.15 (SLC24A5) — i.e., mostly under a ~15% predicted-signal difference between classes:

Gene	log2FC (whole gene span)	log2FC (MANE exons only)
TYR	−0.162	−0.073
TYRP1	−0.072	−0.059
SLC45A2	−0.057	−0.037
MFSD12	−0.049	−0.016
KITLG	−0.048	−0.069
ASIP	−0.038	−0.040
DDB1	−0.009	−0.010
OCA2	+0.004	+0.001
HERC2	+0.007	+0.007
MC1R	+0.079	+0.120
SLC24A5	+0.147	+0.061

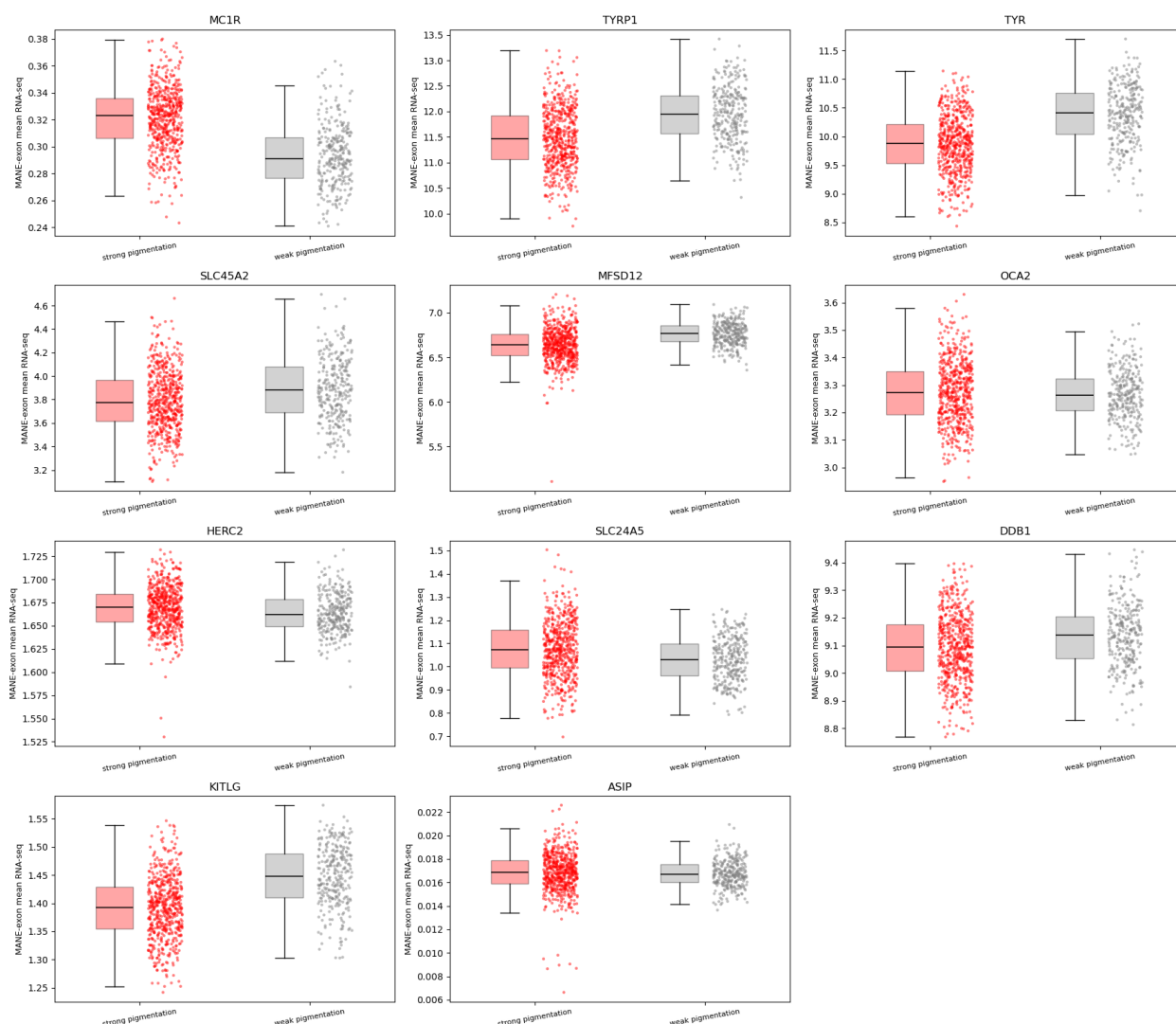
Statistical significance, per-individual (not per-class-mean). A Mann-Whitney U test (rank-based, appropriate for non-normal/zero-inflated expression-like signal) was run per gene on per-individual MANE-exon mean signal, train split only (n=862), with Benjamini-Hochberg FDR correction across the 11 genes tested:

Gene	n strong / n weak	strong mean	weak mean	p-value	FDR q-value	Significant (q<0.05)
MC1R	570 / 292	0.321	0.293	1.2e-48	1.4e-47	Yes
KITLG	570 / 292	1.392	1.447	4.0e-37	2.2e-36	Yes
TYR	570 / 292	9.869	10.380	7.7e-36	2.8e-35	Yes
TYRP1	570 / 292	11.471	11.951	7.3e-27	2.0e-26	Yes
MFSD12	570 / 292	6.638	6.765	3.3e-25	7.3e-25	Yes
SLC24A5	570 / 292	1.074	1.030	2.3e-7	4.2e-7	Yes
SLC45A2	570 / 292	3.788	3.886	1.4e-6	2.2e-6	Yes
HERC2	570 / 292	1.670	1.664	6.0e-5	7.4e-5	Yes
DDB1	570 / 292	9.092	9.134	2.0e-6	2.8e-6	Yes
ASIP	570 / 292	0.0168	0.0167	0.195	0.214	No
OCA2	570 / 292	3.273	3.266	0.382	0.382	No

9 of 11 genes are "significant" — but read this together with the effect-size table above: with n≈862 and per-individual variance, Mann-Whitney reaches astronomically small p-values (down to 1e-48) for fold-changes as small as 1-2%. **This is a textbook large-n statistical-vs-practical-significance situation:** statistical significance here mostly reflects sample size and a population-level systematic shift, not necessarily a biologically large or individually discriminating effect. **ASIP and OCA2 were excluded from the classifiers below** as a result of this test — notably, OCA2, one of the most famous, largest-effect pigmentation genes in the literature (Section 3), shows no detectable difference here. This recurs in Section 9 and Section 11 and is discussed as a specific, telling finding in Section 14.

The box+rain plot below shows exactly what this test operates on — the actual per-individual MANE-exon mean values behind the table above, one panel per gene. The overlap between the two classes' distributions is substantial for every gene, which is the visual counterpart of the "large-n" caveat just made:

Per-individual MANE-exon mean RNA-seq, by pigmentation class, per gene (train split) -- box + rain plot; same values the Mann-Whitney significance test below operates on



6. Model training (04_modeling.ipynb)

Three models were trained, all class-weight-balanced (train split is ~2:1 strong:weak) and evaluated on the same untouched 210-person test split:

1. `mane_exon_model` : `StandardScaler` + `LogisticRegression` , one feature per gene = $\log(\text{MANE-exon mean signal} + 0.001)$, using the **9 significant genes** from Section 5. A deliberately simple linear baseline (no tuning/cross-validation) — the goal was to check whether these 9 genes' signal is separable at all.
2. `gene_span_model` : identical setup, but the feature is the whole-gene-span log-mean instead of exon-restricted — a check of whether restricting to exons was actually helping.
3. `nn_model` (`TinyMLP`): a small 2-layer MLP (32 hidden units, dropout 0.5), fed the full **raw per-position** predicted signal for **all 11 genes** (not just the 9 significant ones), concatenated into one 1,010,926-dimensional vector per individual. **This is an extreme $p \gg n$ regime: 1,010,926 features against 862 training individuals** — flagged explicitly in the notebook as a heavy-overfitting-risk setting to be read as a rough sanity check, not a tuned result. Trained 50 epochs with class-balanced `BCEWithLogitsLoss` ; **train accuracy reached 1.000 by epoch 20** and stayed there.

7. Held-out evaluation (05_evaluation.ipynb)

All three models evaluated on the same 210 held-out individuals (133 strong / 77 weak):

Model	Accuracy	ROC-AUC	Confusion matrix (rows=true, cols=pred; 0=weak, 1=strong)
mane_exon_model (9 scalar features)	0.833	0.925	[[61,16],[19,114]]
gene_span_model (9 scalar features)	0.895	0.965	[[64,13],[9,124]]
nn_model (1,010,926 raw features)	0.990	0.992	[[77,0],[2,131]]

The neural model's near-perfect held-out performance, given it reached **perfect training accuracy in a p>n regime**, is exactly the pattern expected if the model is exploiting broad, highly-redundant population-structure signal that is present at a large fraction of the ~1M input positions (both African and European ancestry leave genome-wide, not just gene-specific, footprints in a densely-sampled 500kb window) — a confound present identically in train and test since both were drawn from the same two population panels. It should not be read as evidence that AlphaGenome has learned fine-grained pigmentation regulatory biology without further, independent validation (see Sections 9 and 14).

Gene coefficients (both LogisticRegression models, sorted by |coefficient|):

Gene	mane_exon_model coef	gene_span_model coef
TYR	-1.067	-0.925
MFSD12	-1.038	-2.159
MC1R	+1.001	+1.268
KITLG	-0.945	-0.673
TYRP1	-0.922	-0.859
SLC45A2	-0.502	-0.523
SLC24A5	+0.195	+0.609
DDB1	-0.190	-0.181
HERC2	+0.167	+0.173

A specific, worth-flagging oddity: for 5 of 9 genes (TYR, MFSD12, KITLG, TYRP1, SLC45A2) — including several of the most canonical melanogenesis-enzyme genes — higher predicted expression is associated with the "weak pigmentation" class in this data, opposite the naive expectation that upregulating melanin-synthesis enzymes would associate with more pigmentation. This is not necessarily wrong (real regulatory biology is often counter-intuitive, and total mRNA level is not the same as protein activity — SLC24A5's well-known causal variant, for example, changes protein function, not necessarily transcript abundance), but combined with the near-chance GTEx concordance result in Section 9, it is at least as consistent with **AlphaGenome's predicted expression direction not being reliable ground truth for this gene panel** as with a genuine, surprising biological finding. This is not resolved by the current analysis and is flagged as an open question in Section 14.

Reference-genome baseline: `nn_model` predicts the unmodified GRCh38 reference genome as "weak pigmentation" with extremely high confidence ($P[\text{strong}]=0.000$, $\text{logit} \approx -1,589$). Every subsequent variant-effect analysis (Sections 8–9) measures a shift from this baseline.

8. Do individual literature variants push predictions the "expected" direction?

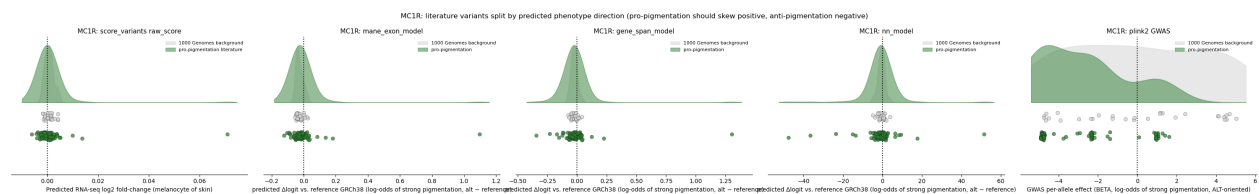
For each of the 6 significant genes with at least one GTEx skin-eQTL in its window (KITLG, MC1R, MFSD12, SLC24A5, TYR, TYRP1 — DDB1, HERC2, and SLC45A2 have **zero** GTEx skin-eQTLs in-window and were explicitly skipped, not silently dropped), up to 100 literature (GTEx eQTL) variants and 30 random 1000-Genomes background variants were scored with AlphaGenome across **5 methods**: the model-independent `score_variants` raw score (predicted RNA-seq log2FC), the two LogisticRegression models' delta-logit relative to the reference genome, the neural model's delta-logit, and (added in a later iteration, see Section 11) a plink2 GWAS per-allele effect size.

Each literature variant's expected direction ("pro-" vs. "anti-pigmentation") combines **two** things: the variant's own GTEx-measured direction ("enhance" or "diminish" expression — this varies per variant, since a gene can have multiple independent eQTLs, some raising and some lowering expression) and the **gene's own trained coefficient sign** (Section 7 — one fixed value per gene, which fixes which combination of enhance/diminish counts as "pro" for that gene). A variant is "pro-pigmentation" if its measured direction agrees with the gene's coefficient-sign convention, "anti-pigmentation" if it disagrees — so a gene whose eQTLs are all one direction gets a 100%/0% split (not independent evidence of anything, close to guaranteed by construction), while a gene with genuinely bidirectional eQTLs splits accordingly (TYR, 32/32, below). Using the gene's own coefficient sign at all — rather than an independently curated table — is what the notebook flags as making the check partially circular for `mane_exon_model`'s own logit (a "correct-side" result there partly just confirms internal consistency) but not for the other 4 scoring methods.

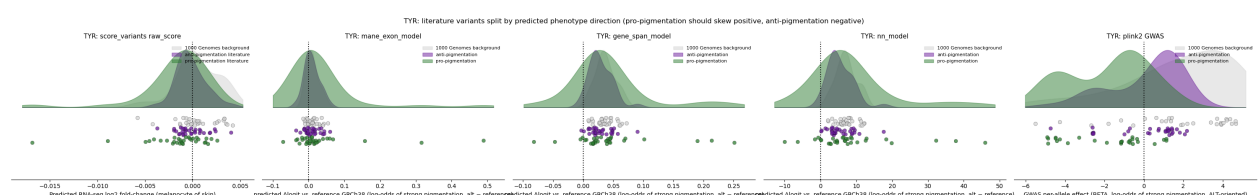
Gene	Literature variants (GTEx eQTLs)	Pro-pigmentation	Anti-pigmentation
MC1R	100	100	0
KITLG	59	0	59
TYRP1	35	32	3
TYR	64	32	32
MFSD12	13	11	2
SLC24A5	2	2	0

MC1R and KITLG happen to have all their GTEx eQTLs pointing one expression direction, so the split is 100%/0%; TYR's eQTLs are genuinely bidirectional (32 enhance, 32 diminish), giving an exact 50/50 split.

Two real examples of the resulting 5-panel rain plots (all 6 genes' plots are in `05_evaluation.ipynb`; these two illustrate the range of outcomes). MC1R (100/0 pro-pigmentation, by construction — see above) shows its literature variants sitting almost entirely on one side of zero across every panel, background included:



TYR (32 pro / 32 anti) is more representative of the panel as a whole — pro- and anti-pigmentation groups substantially overlap in most panels, with the clearest separation only showing up in the GWAS BETA panel on the right:



9. Does AlphaGenome's predicted direction agree with GTEx's independently measured direction?

This is the most important internal validity check in the whole project, and it is a sobering result. Unlike Section 8 (which derives "expected direction" from this study's own trained coefficients), this check compares AlphaGenome's **model-independent** `score_variants` raw score sign against **GTEx's own measured NES sign** — two genuinely independent signed quantities, both about the same thing (does this allele raise or lower this gene's expression).

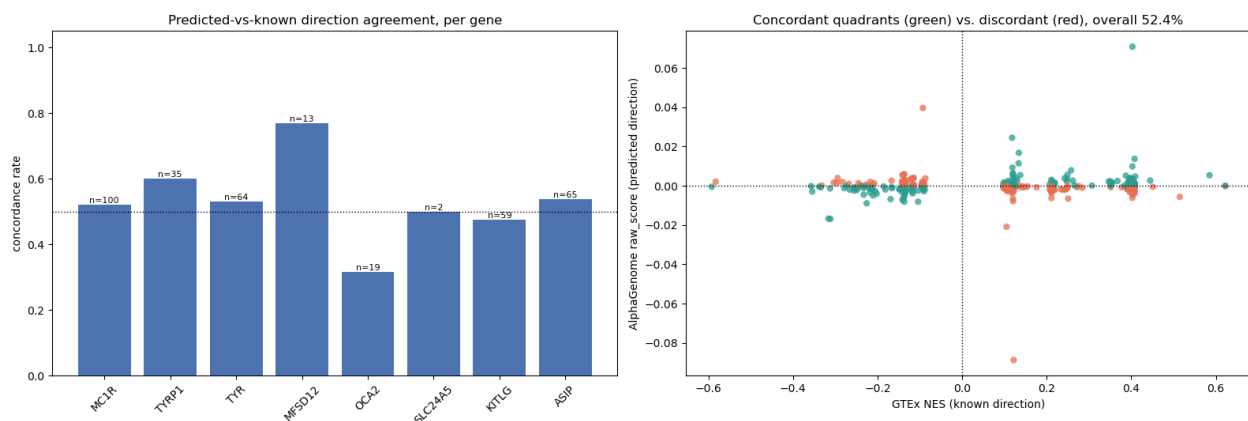
Across all 11 genes (up to 100 literature variants/gene, 357 variants total across the 8 genes with any GTEx eQTLs):

Overall concordance: 52.4% (187/357) — the notebook's own automated check flags this explicitly: "WARNING: concordance is near chance (50%)."

Gene	n variants	Concordance rate
MFSD12	13	76.9%
TYRP1	35	60.0%
TYR	64	53.1%
ASIP	65	53.8%
MC1R	100	52.0%
SLC24A5	2	50.0%
KITLG	59	47.5%
OCA2	19	31.6% (below chance)
SLC45A2 / HERC2 / DDB1	—	no GTEx eQTLs in window

Only MFSD12 (n=13) shows a concordance rate that looks clearly better than chance; OCA2 is below chance. This means that, on this gene panel, AlphaGenome's own predicted direction of effect on gene expression does not reliably track an independent, real measurement of the same quantity (GTEx eQTL direction). Notably, **OCA2 is again the outlier** — the same gene that showed no significant population difference in Section 5 also shows below-chance GTEx concordance here, a consistent (if negative) signal across two independent parts of the pipeline, discussed further in Section 14.

Left: per-gene concordance rate (dotted line = chance). Right: every scored variant, GTEx's known direction (x-axis) against AlphaGenome's predicted direction (y-axis) — green points agree, red points disagree; the near-even mix of colors on both sides of zero is the visual signature of the chance-level result:



10. Interpretability: what does the neural model actually attend to?

(06_interpretability.ipynb)

DeepLIFT (via `captum`) attributions were computed for `nn_model` on all 210 held-out test individuals, using a zero/population-mean baseline (meaningful because inputs were `StandardScaler`-transformed — zero is the training-set mean). The completeness property (`sum(attribution) == output(x) - output(baseline)`) was verified on 5 example individuals and held to 3+ decimal places, confirming the attributions are computed correctly.

Important scope caveat, stated in the notebook and worth repeating here: the model's input is not DNA sequence — it's AlphaGenome's own predicted RNA-seq signal per position. So this measures "which positions' predicted expression level drive the classification," not which DNA bases/motifs do. It is one step removed from sequence-level interpretability.

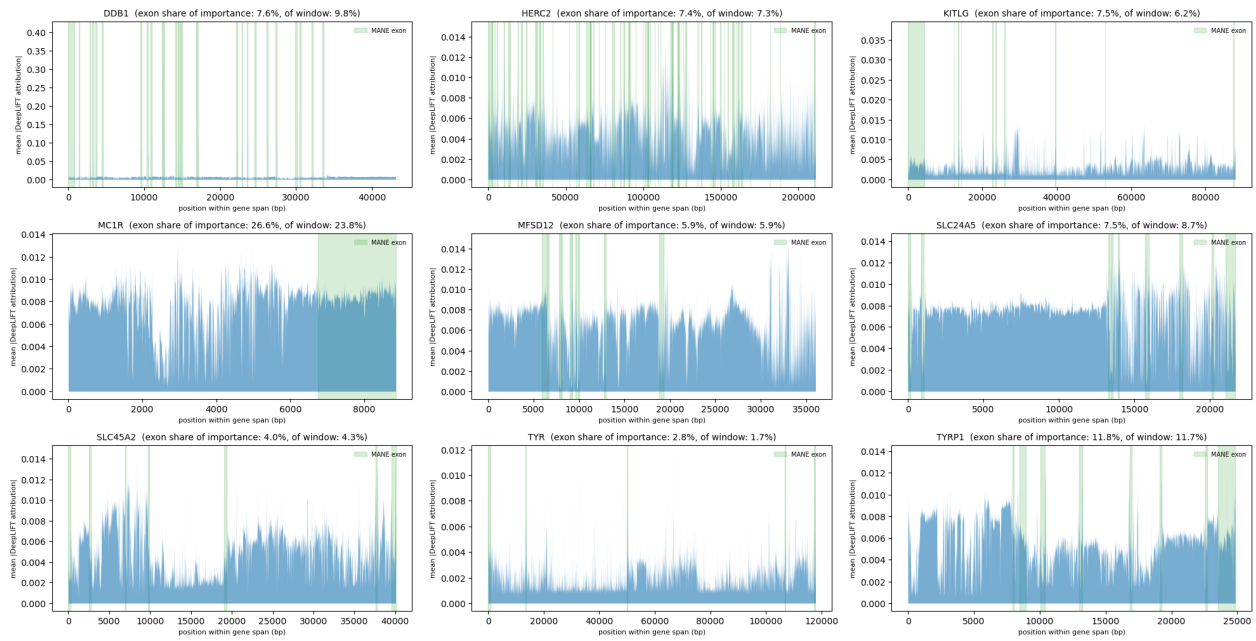
Per-gene exon-importance enrichment (mean $|\text{attribution}|$ concentrated in MANE Select exons, relative to what exons' share of the window would predict by chance — 1.0 = proportional, no exon enrichment):

Gene	Exon share of window	Exon share of importance	Enrichment
TYR	1.7%	2.8%	1.60x
KITLG	6.2%	7.5%	1.21x
MC1R	23.8%	26.6%	1.11x
HERC2	7.3%	7.4%	1.02x
TYRP1	11.7%	11.8%	1.01x
MFSD12	5.9%	5.9%	1.00x
SLC45A2	4.3%	4.0%	0.92x
SLC24A5	8.7%	7.5%	0.87x
DDB1	9.8%	7.6%	0.77x

This is a weak and mixed signal. Only TYR shows importance clearly concentrated above its exon share; three genes (SLC45A2, SLC24A5, DDB1) show importance below their exon share (i.e., more important signal sits in introns/flanking sequence than exons for those genes). Most genes sit close to 1.0x — i.e., importance is roughly proportional to how much of the input window is exonic, which is close to what you'd expect even from a model with no learned exon-specific structure at all. This does not strongly support the hypothesis that the model is preferentially using exonic (splicing/expression-relevant) signal over intronic signal.

Blue = mean $|\text{DeepLIFT attribution}|$ per position (held-out test set); green shading = MANE Select exons. Visually, importance is spread broadly across each window rather than sharply picking out the green-shaded exons — the same "weak, mixed" pattern the enrichment table quantifies:

Per-position DeepLIFT importance (blue) vs. MANE Select exon structure (green shading), per gene -- held-out test set



11. Classical-statistics baseline: plink2 GWAS (07_gwas_baseline.ipynb)

To answer "does AlphaGenome add anything over a standard, well-understood genetic-association method?", a real plink2 GWAS was built on the same cohort/phenotype/candidate-gene panel, with genuine QC (not a strawman): MAF/missingness filtering, LD-pruning, relatedness handling, HWE filtering in controls, and Bonferroni correction scaled to the actual number of tested variants (not a genome-wide 5×10^{-8} threshold, which would be the wrong frame for an 11-gene candidate-region test).

Design note, and a real mid-course correction worth mentioning: the standard approach to relatedness (--king-cutoff , genotype-based kinship) was tried first and produced an implausible result — it flagged 69% of the cohort as "related" when computed from markers confined to one 524kb candidate gene, because a marker set restricted to the most population-differentiated loci in the genome conflates "same population" with "genetically related," which genotype-based kinship methods assume it does not. This was caught by inspection (69% relatedness is not plausible for this cohort) and fixed by switching to **pedigree-based deduplication** (one individual per 1000 Genomes family, using the database's own published family IDs) — both more correct for this dataset and simpler.

Deliberately, **no ancestry-PC adjustment was applied** — with a population-membership phenotype, ancestry PCs would be almost perfectly collinear with the outcome being tested, so "correcting" for them would remove the signal being measured, not a confound. This mirrors the same limitation already present on the AlphaGenome side (Section 2) rather than hiding it, which is what makes this a fair baseline instead of a strawman comparison.

QC funnel (from 1,072 individuals / 123,895 variants across the 11-gene merged panel):

Step	Samples	Variants
Raw	1,072	123,895
After $MAF \geq 1\%$ / genotype-missingness $\leq 10\%$ / sample-missingness $\leq 10\%$	1,072	26,966
LD-pruned (informational, for an alternate Bonferroni basis)	—	2,869
Pedigree-deduplicated (one per family)	588	—
HWE-in-controls ($p > 1e-6$)	—	26,931
Final analysis set	588	26,931

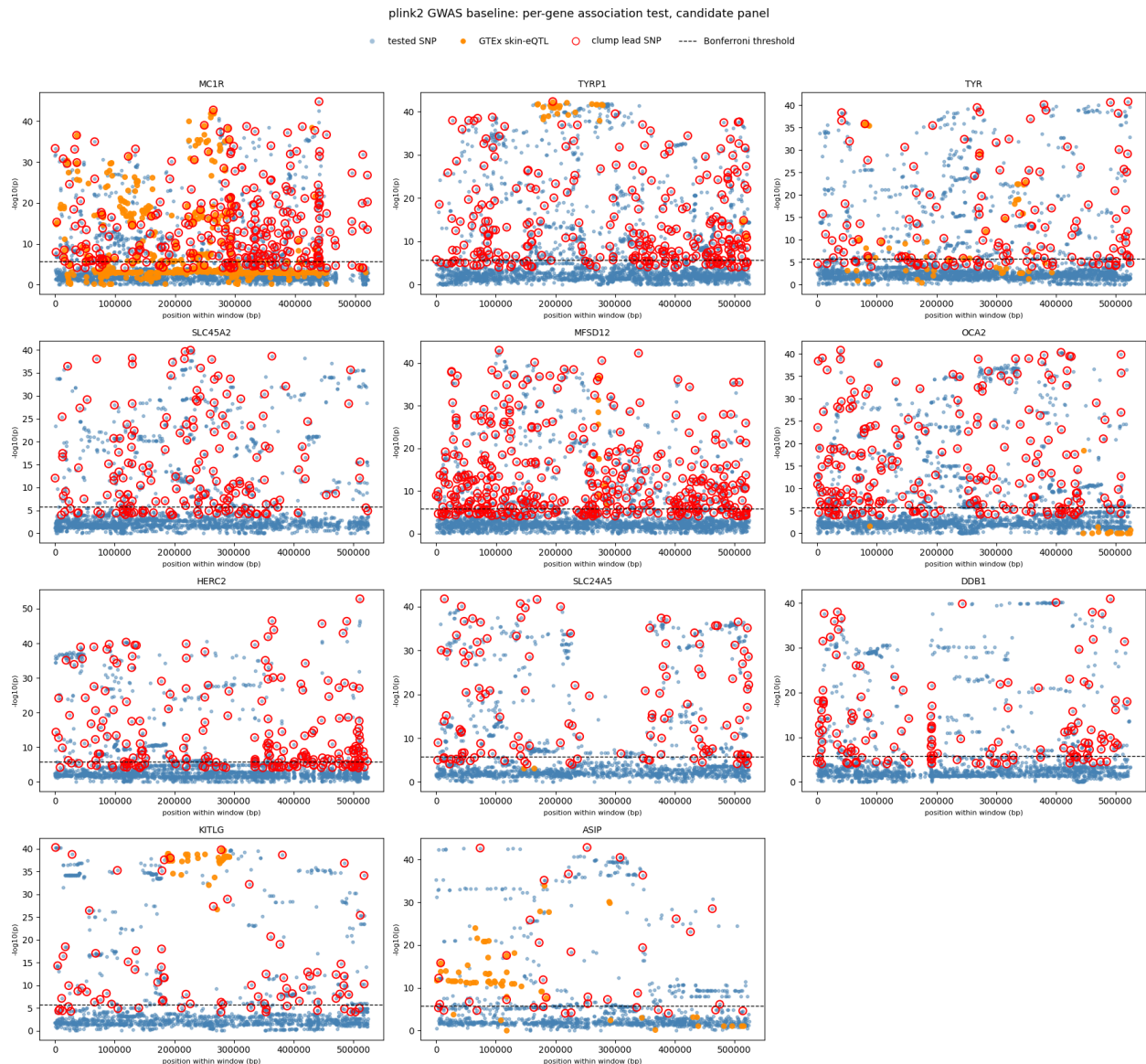
Association (logistic regression, sex-covaried only, Firth-fallback for rare variants):

Gene	Tested variants	Bonferroni-significant ($p < 1.86e-6$)	% significant
MC1R	3,542	987	27.9%
TYRP1	3,269	1,071	32.8%
TYR	2,986	692	23.2%
SLC45A2	2,396	636	26.5%
MFSD12	3,241	863	26.6%
OCA2	3,071	956	31.1%
HERC2	2,516	568	22.6%
SLC24A5	1,943	584	30.1%
DDB1	1,986	515	25.9%
KITLG	1,772	491	27.7%
ASIP	1,398	397	28.4%
Total	26,931	7,404 (27.5%)	

27.5% of all tested variants across the panel clear a stringent Bonferroni threshold — not the pathological 100% that would signal a pure statistical artifact, but still a very high rate, consistent with the expected signature of broad AFR/EUR allele-frequency differentiation across these regions (Section 2), not fine-mapped single-variant causality. LD-based clumping (index $p \leq 1e-4$, $r^2 > 0.5$, within 250kb) collapses this down to **2,040 independent lead signals** across the panel; the single strongest hit in the entire panel is in HERC2 (chr15:28,465,010, $p = 1.5 \times 10^{-53}$), immediately followed by MC1R (chr16:90,094,439, $p = 1.6 \times 10^{-45}$) — exactly the loci with the best-established, largest-effect allele-frequency differences between African and European populations in the population-genetics literature. OCA2 itself (a distinct gene from HERC2, though their windows overlap — Section 3) has its own strongest hit at chr15:27,703,972, $p = 1.5 \times 10^{-41}$ — independently confirmed, not just inherited from its neighbor. This is a real, if unsurprising, positive control: the GWAS baseline correctly recovers the strongest known signal in the panel using nothing but 60-year-old logistic regression.

The pointed comparison for the "what does AlphaGenome add" question: OCA2 — which showed no significant population difference in the AlphaGenome-based expression test (Section 5) and below-chance GTEx concordance (Section 9) — is the **second-most GWAS-significant gene in the entire panel by Bonferroni fraction (31.1%)**, with its own strongest single-variant association at $p = 1.5 \times 10^{-41}$. Genotype-level classical statistics detects OCA2's population-differentiation signal clearly and strongly; the AlphaGenome-predicted-expression pipeline does not detect it at all. This is the clearest concrete evidence in this report that the AlphaGenome-expression intermediate step is not simply a strictly-more-informative superset of what classical genotype-based analysis already finds — at least for this gene, it appears to lose signal that a much simpler method retains. (GWAS's coverage of the literature/background variant sets used in Section 8's rain plots was also checked directly: 85-100% of those variants survived GWAS QC and could be cross-compared, confirming this isn't an artifact of non-overlapping variant sets.)

Every tested variant, all 11 genes, position within window (x) vs. $-\log_{10}(p)$ (y): blue = tested SNP, orange = a GTEx skin-eQTL, red circle = an independent LD-clumped lead SNP, dashed line = Bonferroni threshold. Note how much of each panel — OCA2's and HERC2's especially — sits above the line:



12. What was done well (methodological strengths worth preserving)

- **Held-out evaluation on a genuinely unseen, family-disjoint, population-stratified test split**, used consistently across all three models and both evaluation approaches.
- **Gene selection was corrected once shown to be unsound**, rather than the flawed automated ranking being kept or quietly patched — the failure mode (LD-block-size confound) was diagnosed concretely, not hand-waved.
- **FDR correction** (Benjamini-Hochberg) applied to the 11-gene significance test, avoiding uncorrected multiple-comparisons inflation.
- **A real, non-strawman classical GWAS baseline** was built specifically to stress-test the AlphaGenome results, with genuine QC (MAF/missingness/HWE/relatedness/LD), a Bonferroni threshold correctly scaled to a candidate-region (not genome-wide) test, and a documented mid-course correction when the first relatedness-handling approach produced an implausible result. Multiple-testing correction was reported at two different rigor levels (raw-count and LD-pruned-count Bonferroni) rather than picking the more favorable one.
- **An independent concordance check against GTEx** was built and its result (52.4%, near chance) was reported as-is, including an automated in-notebook warning, rather than only reporting the more favorable per-gene rain-plot direction checks (Section 8) that are partially circular by construction.

- **Missing data handled by explicit reporting, not silent omission**, throughout: genes with zero GTEx eQTLs, variants that didn't survive GWAS QC, near-100% Bonferroni-significance flagged as a named warning condition — this is unusually disciplined bookkeeping for an exploratory analysis.
- **Interpretability (DeepLIFT) was verified for correctness** (completeness property checked numerically) before its results were interpreted.

13. Threats to validity, collected in one place

1. **Phenotype is a population-membership proxy, not a measured trait** (Section 2) — the overarching confound behind everything else in this list.
2. **The neural model operates in an extreme $p \gg n$ regime** (1,010,926 features, 862 training individuals) and reaches perfect training accuracy — a classic overfitting/shortcut-learning signature, made worse (not better) by the population confound, since a shortcut ("does this look African or European") is available at enormous redundancy across the input.
3. **GTEx concordance is at chance overall (52.4%) and below chance for OCA2 (31.6%)** — the one truly independent validation check built in this project did not confirm AlphaGenome's predicted-direction reliability on this gene panel.
4. **Several gene coefficients point the biologically counter-intuitive direction** (Section 7), unresolved and plausibly linked to point 3.
5. **A plain classical GWAS baseline recovers comparable-or-stronger "significance"** using the same confounded phenotype and simple logistic regression, and in OCA2's case, recovers a strong signal that the AlphaGenome-based approach misses entirely (Section 11) — there is currently no result in this project demonstrating that AlphaGenome adds information over this baseline.
6. **OCA2/HERC2 feature correlation** (45% window overlap) complicates independent interpretation of their individual coefficients/GWAS hits.
7. **No true external replication** — all results are internal to one 1000 Genomes cohort split; nothing here has been checked against an independent dataset or a cohort with a real measured phenotype.
8. **DeepLIFT exon-enrichment signal is weak and inconsistent across genes** (0.77x-1.60x, mostly close to 1.0x) — does not strongly support "the model learned exon-specific regulatory biology" over "importance roughly tracks how much of the window is exonic by construction."
9. Minor: one stale sentence survives in `04_modeling.ipynb`'s markdown (a leftover reference to "TCHH" as one of the two excluded genes, from an earlier iteration of the panel) — the actual, verified-correct excluded genes are ASIP and OCA2 (Section 5). Worth a one-line fix, does not affect any number in this report (all numbers here were pulled from executed cell outputs, not from that stale prose).

14. Is this publishable? What would it take?

As currently framed — "AlphaGenome predicts individual pigmentation status / uncovers pigmentation regulatory biology from genotype" — this is not yet publishable. The central claim would need to survive the confound in point 1 above, and right now two of the project's own internal checks (GTEx concordance, the GWAS baseline) are more consistent with "the models are detecting population structure broadly" than with "the models are detecting pigmentation-specific regulatory signal." That is a defensible, useful thing to have discovered through careful analysis — but it is a different (and less exciting) paper than the one the project set out to write.

Two realistic paths forward:

Path A — fix the phenotype, keep the pipeline. Replace the population-membership proxy with a real measured pigmentation phenotype in a cohort with matched genotypes — e.g., a cohort with quantitative skin reflectance/melanin index, or a large biobank with self-reported skin-tone / tanning-response phenotypes and genotypes from a single, non-stratified (or admixed, with proper local-ancestry handling) population, so that "high vs. low pigmentation" is no longer synonymous with "African vs. European ancestry." This is the highest-leverage single change: essentially all of the machinery already built (per-individual consensus prediction, the two scalar classifiers, the neural model, DeepLIFT, the GWAS baseline, the GTEx concordance check) is directly reusable once the phenotype is real. This is also the only path that would let a strong result be interpreted as genuine pigmentation biology rather than ancestry classification.

Path B — reframe as a methods/tooling contribution, explicit that the current cohort's classification task is dominated by population structure, and report that honestly as a finding (e.g., "AlphaGenome-based expression features, like genotype-based methods, are dominated by population structure when a population-confounded phenotype proxy is used; a rigorous baseline comparison and independent concordance check are necessary and were often absent from prior similar work"). This is a legitimate, if more modest, contribution — the GWAS-baseline pipeline, the GTEx concordance methodology, and the DeepLIFT interpretability pipeline are all reusable, well-documented pieces of infrastructure regardless of which phenotype is plugged in.

Before either path, three concrete, comparatively cheap next steps would materially strengthen whichever direction is chosen:

1. **Investigate the near-chance GTEx concordance directly** — is it a sign-convention bug somewhere in the pipeline (worth ruling out explicitly with a few hand-checked variants against raw AlphaGenome/GTEx output), or a real negative result about `score_variants` ' reliability on this gene panel? This is the single highest-priority open question, since it undercuts every variant-level claim in Sections 1 and 8, and would need to be resolved (or the claims scoped down) before any variant-direction result is presented externally.
2. **Formalize the AlphaGenome-vs-GWAS comparison quantitatively**, not just visually (Section 11's 5-panel rain plots exist, but there is no summary statistic yet answering "does AlphaGenome correctly identify/rank the GWAS-significant variants better than chance, and does it add resolution within GWAS's LD-clumped blocks that GWAS itself cannot provide").
3. **Re-run the modeling pipeline with an ancestry-neutral negative control** (e.g., a matched set of non-pigmentation genes, or the same 11 genes but with population labels permuted) to establish what "chance-level, structure-only" performance looks like in this exact pipeline — this would make points 2, 3, and 5 in Section 13 falsifiable rather than argued qualitatively.

15. Where everything lives

Notebook	Role
01_overview_and_vep.ipynb	Setup, gene-panel selection & rationale, family-aware split, classical VEP, MFSD12 deep-dive
02_individual_predictions.ipynb	Per-individual diploid consensus RNA-seq generation (bulk AlphaGenome calls)
03_population_analysis.ipynb	Population-level mean comparison, MANE-exon restriction, box+rain plots, significance test
04_modeling.ipynb	Trains and persists the 3 classifiers
05_evaluation.ipynb	Held-out evaluation, coefficient inspection, literature-variant rain plots (5 scoring methods incl. GWAS), GTEx concordance check
06_interpretability.ipynb	DeepLIFT attribution vs. MANE-exon structure
07_gwas_baseline.ipynb	Standalone plink2 GWAS baseline: QC funnel, association, Bonferroni, LD-clumping

All cached intermediate artifacts (predictions, models, GWAS outputs) live under `notebooks/.cache/` (gitignored); source data under `notebooks/data/` (gitignored, rebuildable via `notebooks/utis/prepare_data.py`).